

Question_1:

Code:

```
1 // Teddy has n cities and m bidirectional roads, each requiring some to
2 // Choose minimum-cost tokens so every city can reach every other city.
3 #include <iostream>
4 #include <vector>
5 using namespace std;
6
7 #define MAXN 100005
8 struct Edge {
9     int u, v, mask;
10 };
11 vector<int> parent(MAXN);
12 long long cost[60];
13 Edge edges[MAXN];
14
15 int find(int i) {
16     if (parent[i] == i)
17         return i;
18     return parent[i] = find(parent[i]);
19 }
20 int main() {
21     int n, m, k;
22     cin >> n >> m >> k;
23
24     for (int i = 0; i < k; i++)
25         cin >> cost[i];
26
27     for (int i = 0; i < m; i++) {
28         int u, v, l;
29         cin >> u >> v >> l;
30
31         int mask = 0;
32         for (int j = 0; j < l; j++) {
33             int token;
34             cin >> token;
35             token--;
36             mask |= (1 << token);
37         }
38         edges[i] = {u, v, mask};
```

```
39     }
40     long long minCost = -1;
41
42     for (int mask = 0; mask < (1 << k); mask++) {
43
44         for (int i = 1; i <= n; i++)
45             parent[i] = i;
46
47         int components = n;
48         for (int i = 0; i < m; i++) {
49             if ((edges[i].mask & mask) == edges[i].mask) {
50
51                 int ru = find(edges[i].u);
52                 int rv = find(edges[i].v);
53
54                 if (ru != rv) {
55                     parent[ru] = rv;
56                     components--;
57                 }
58             }
59         }
60         if (components == 1) {
61             long long currentCost = 0;
62
63             for (int i = 0; i < k; i++)
64                 if ((mask >> i) & 1)
65                     currentCost += cost[i];
66
67             if (minCost == -1 || currentCost < minCost)
68                 minCost = currentCost;
69         }
70     }
71     cout << minCost << endl;
72     return 0;
73 }
```

Output:

Test Case_1:

```
3 3 4
1 2 5 10
1 2 2 1 2
1 3 1 3
2 3 1 4
8
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
3 3 4
11 21 15 10
1 21 2 11 2
1 13 1 3
2 31 1 4
25
PS C:\Users\Prabin Yadav\
```